Hospital Emergency Room Performance Analysis

An operational analytics case study examining patient wait times, satisfaction drivers, admission patterns, and demographics across 10,100 Emergency Room visits.

The Business Question

A hospital needed to understand where patient experience was breaking down and how to improve operational efficiency. Four questions drove the analysis:
- What drives long patient wait times?
- What drives low patient satisfaction?
- Which departments face the highest operational pressure?
- Who is coming to the ER, and when?

The Findings

Wait times vary across departments and physicians.

The average patient wait time was 31 minutes, with observed waits ranging from 1 to 199 minutes. Gastroenterology recorded the highest average wait among departments. The top 10 physicians by average wait time ranged from approximately 32 to 36 minutes, compared with the hospital average of 31 minutes. Mark Smith recorded the highest physician average at 36.28 minutes. Overall, 8.81% of patients waited more than 60 minutes.

Patient satisfaction also varies by department.

Gastroenterology recorded the highest average satisfaction score at 5.75 out of 10, while Physiotherapy recorded the lowest at 4.56. Patients waiting 61-120 minutes reported an average satisfaction score of 5.37, compared with 4.91 among patients waiting 0-15 minutes. In this dataset, longer waits were not consistently associated with lower satisfaction. The available data does not explain why, so other factors affecting patient experience would need further investigation.

ER demand shows clear differences by month, day, and age group.

May recorded the highest monthly volume at 874 visits, followed closely by December at 867 and March and August at 861 each. Monday recorded the highest daily visit volume. Patients aged 31–45 formed the largest age group, accounting for 2,778 visits.

The Recommendation

- Review departments and physicians with above-average wait times to identify workflow, scheduling, or case-mix differences
- Investigate the satisfaction gap in Physiotherapy and other lower-scoring departments before deciding on corrective action
- Use monthly and day-of-week demand patterns to support staffing and capacity planning
- Examine the needs and visit patterns of the 31–45 age group, the largest patient segment in the dataset
- Review low physician satisfaction scores alongside patient volume, department, wait time, and available patient characteristics before identifying coaching needs

What This Project Demonstrates

Skill	Where Used	Evidence
Excel & Power Query	Data import and cleaning	Standardized 8 categorical fields; removed 30 invalid records; engineered 4 date features
Pivot Tables	All four dashboards	Built structured pivot tables across wait time, satisfaction, demographics, and physician dimensions
Data Cleaning	Pre-analysis audit	Documented and resolved missing values, invalid records, and inconsistent categories
Star Schema Modeling	Data preparation	Built 1 Fact table + 4 Dimension tables before dashboarding
Dashboard Design	All four pages	Linked dashboards with KPI cards, bar charts, donut charts, and line charts
Business Analysis	Entire project	Investigated four operational questions and linked the findings to practical recommendations
Data Storytelling	Executive summary	Structured as question - data - finding - recommendation

Data Preparation Summary

The raw dataset (10,130 records) required significant cleaning before analysis. Key issues identified and resolved:

- 5,523 missing Department values replaced with "Unknown" to preserve row count
- 7,088 missing Satisfaction Scores retained as blank, absent feedback is not the same as poor satisfaction
- 30 invalid wait-time records removed (16 negative, 14 placeholder values of 999 minutes)
- Gender standardized from M / F / Male / Female / "Femaleemaleemale" to Male / Female
- Triage levels standardized from 1 / L1 / ESI 1 / Level 1 to Level 1–5
- Shift standardized from Morning / MORNING / Morning Shift to Morning / Afternoon / Night
- Date features engineered: Visit Year, Month, Day of Week, Hour

Final cleaned dataset: 10,100 records.
Full cleaning documentation in `Hospital_Emergency_Audit.pdf`.

Files in This Repository

- Hospital_ER_Executive_Summary.docx: Full written case study with findings and recommendations
- Hospital_ER_Analysis.pdf: Presentation deck (8 slides)
- Hospital_Emergency_Audit.pdf: Data quality audit report documenting all cleaning steps
- Hospital_Projects.xlsx: Excel workbook with four dashboards and underlying pivot table

Tools Used

Excel: Dashboard design with KPI cards, slicers, bar charts, donut charts, and line charts
Power Query: Data import, standardization of 8 categorical fields, date feature engineering
Pivot Tables: Structured aggregations across departments, physicians, shifts, triage levels, demographics, and time dimensions

About Me

Victoria Johnson | Revenue Operations Analyst
Remote US/UK EST

4+ years of B2B sales experience in US building materials and OEM manufacturing markets, transitioning into Revenue Operations analytics.

Email: Johnsonfisayo090@gmail.com
Phone: +1 (915) 209-5153
GitHub: github.com/Vickytoria-lab